

Clustering

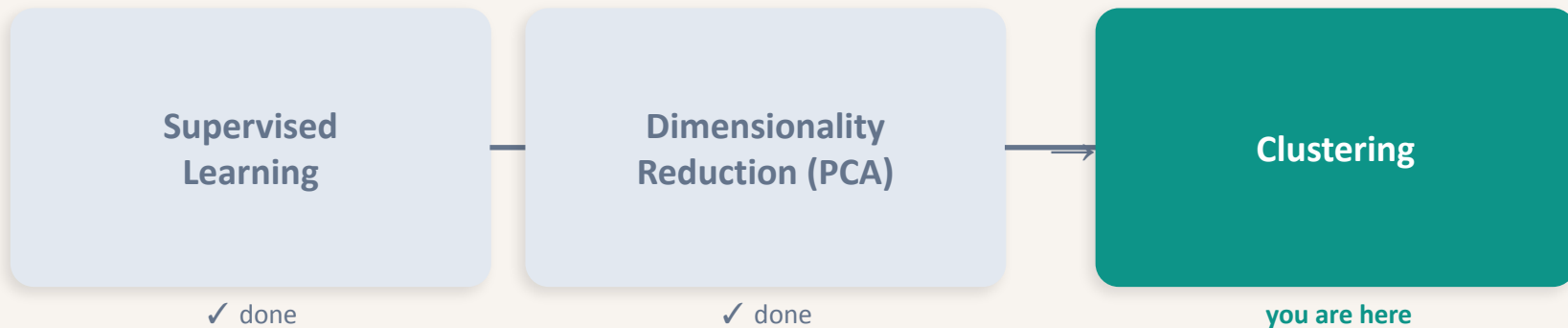
Finding hidden structure without labels

QLSC 612 · Machine Learning Module · Part II

by Mohammad Torabi



Where are we in the ML journey?



Today's goal: learn how to discover hidden groups in data — without any labels.

Today's agenda

-  Quick PCA recap + why it matters for clustering
-  What is clustering? K-means deep dive
-  Real example from the ORIGAMI lab: motor task fMRI
-  Evaluation: internal vs external validation + why labels still matter

Quick PCA Recap — Why It Matters Here

PCA in one sentence

- Find new axes (principal components) that capture the most variance in data
- Result: fewer dimensions, most signal kept
- Transforms $X (n \times p) \rightarrow W (n \times m)$ where $k \ll m$

Why run PCA before clustering?

- Curse of dimensionality: Euclidean distances become meaningless in high-D
- Noise removal: PCA keeps signal, drops noise dimensions
- Speed: K-means on 20 PCs is far faster than on thousands of features
- Visualization: 2D PCA plot is the first thing you should look at

 In today's example: each time point has ~thousands of connectivity features $\rightarrow$ PCA is essential before K-means



What is Clustering?

Clustering: Grouping Without Labels

Supervised Learning

- ✓ Labels are known (y)
- ✓ Goal: learn a mapping $X \rightarrow y$
- ✓ Example: predict diagnosis

VS

Unsupervised: Clustering

- ✗ No labels available — or labels are incomplete
- ✓ Goal: Find structure in X
→ group similar samples by measuring distance between them in feature space
- ✓ Intuition: nearby points → same cluster; distant points → different clusters

Real-world clustering examples:



Brain states
from fMRI time series



Tumour subtypes
from gene expression



Neural patterns
during motor task



Patient groups
from clinical data

What Makes a Good Cluster?

Within-cluster cohesion

Points inside a cluster should be close to each other
(small intra-cluster distance)

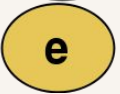
Between-cluster separation

Different clusters should be far apart from each other
(large inter-cluster distance)

 **Key insight: clustering result depends on (a) your distance metric and (b) number of clusters k — these are choices YOU make!**

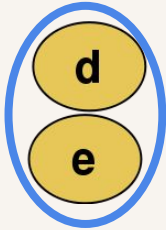
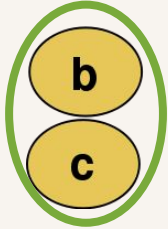
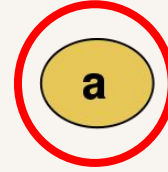
Discussion: If you had brain connectivity data during a motor task, what groups of time points might you expect to find?

Which Samples will Cluster Together?

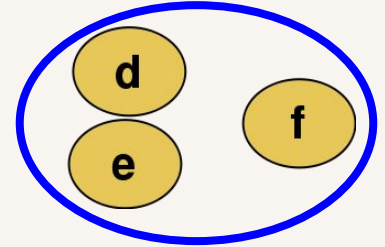
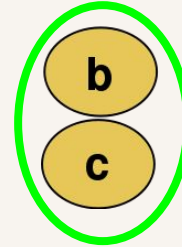


Sample space

Which Samples will Cluster Together?



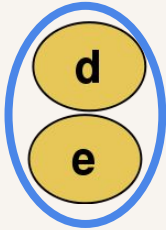
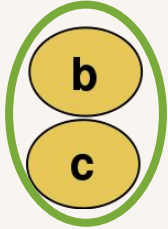
Sample space



Sample space

Discussion: Which clustering is more accurate ?

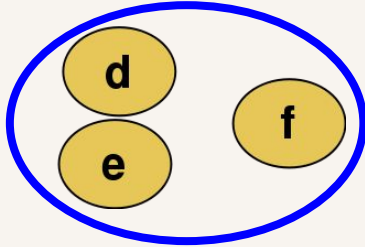
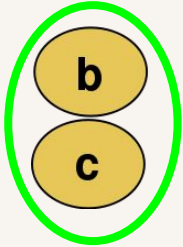
Which Samples will Cluster Together?



Sample space

Discussion: Which samples will cluster together if $n_clusters=4$?

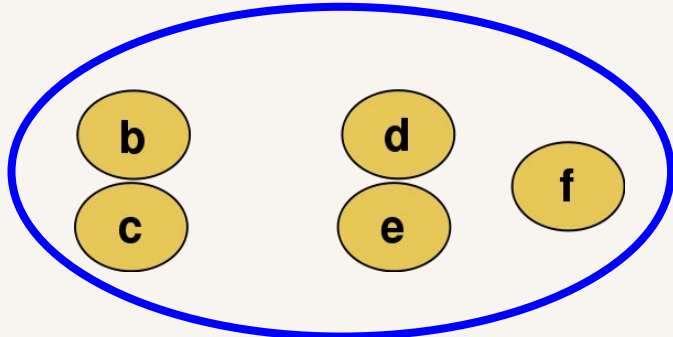
Which Samples will Cluster Together?



Sample space

Discussion: Which samples will cluster together if $n_clusters=3$?

Which Samples will Cluster Together?



Sample space

Discussion: Which samples will cluster together if $n_clusters=2$?

Clustering is in the Eye of the Beholder

Different algorithms — same data — very different results

K-Means

✓ Fast, scalable

⚠ Assumes spherical clusters

Agglomerative

✓ Any shape, no k needed

⚠ Slow on large data

DBSCAN

✓ Finds arbitrarily shaped clusters

⚠ Sensitive to density params

Gaussian Mixture

✓ Soft assignments

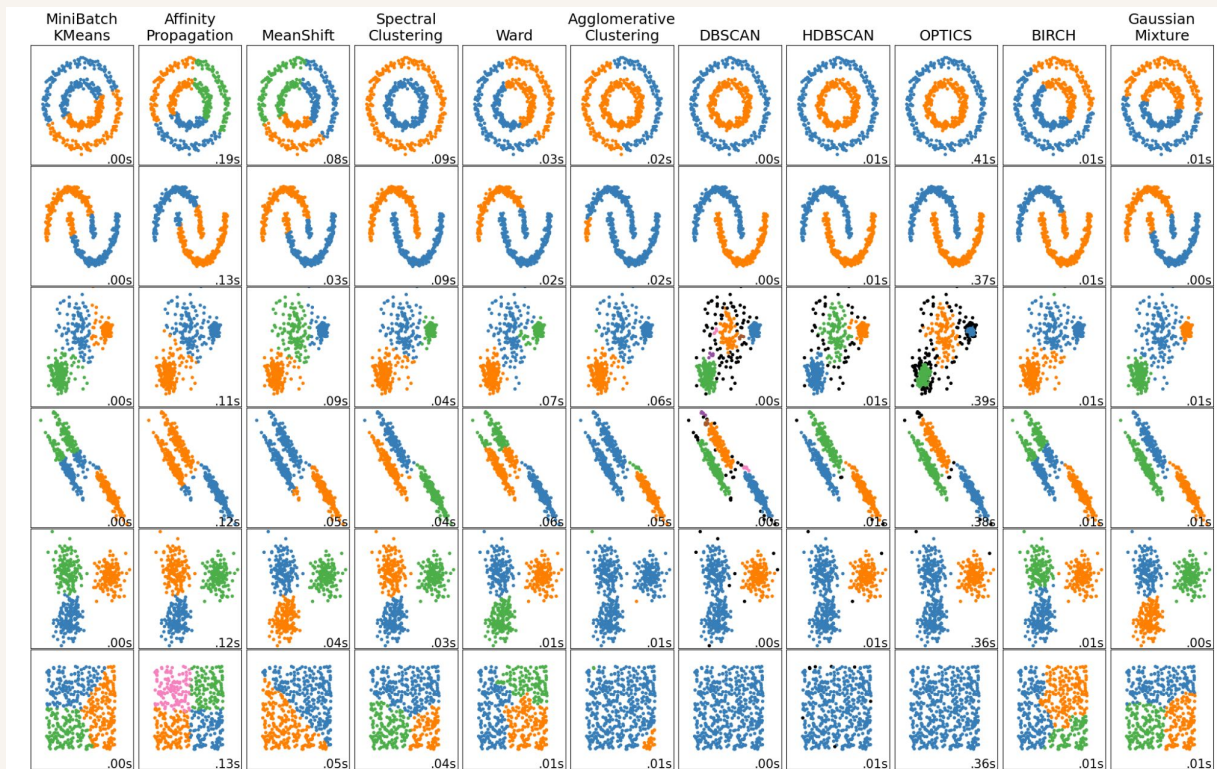
⚠ Needs distribution assumption



In this course we focus on K-Means — most widely used, especially with PCA pre-processing

Clustering is in the Eye of the Beholder

Different algorithms — same data — very different results





K-Means Clustering

K-Means: The Algorithm

1

Initialise

Randomly place k centroids
in the feature space

2

Assign

Each point joins the cluster
of its nearest centroid

3

Update

Recompute each centroid
as the mean of its members

4

Repeat

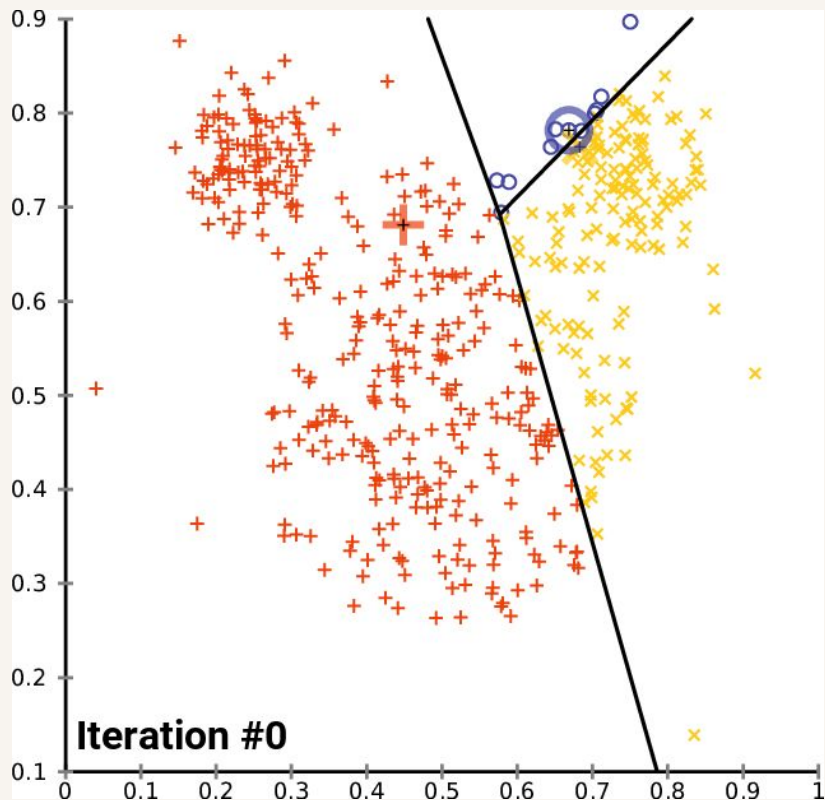
Repeat steps 2-3 until
assignments stop changing

Objective: minimise within-cluster sum of squared distances (inertia)

$$\text{Inertia} = \sum \min \|x_i - \mu_j\|^2$$

where x_i = data point, μ_j = centroid of cluster j

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K-Means: What Can Go Wrong?



Random init sensitivity

Different random seeds → different results.

Fix: run multiple times (sklearn default: `n_init=10`)



Must specify k upfront

You need to decide the number of clusters before running.

Fix: try multiple k values, use metrics to choose



Assumes spherical clusters

Struggles with elongated or irregular shapes.

Fix: transform features; consider other algorithms



Sensitive to outliers

One extreme point can pull a centroid far away.

Fix: remove outliers or use robust alternatives



Euclidean distance only

Not ideal for all data types (e.g. correlation structure).

Fix: run PCA first or use cosine distance variants



High-dim spaces

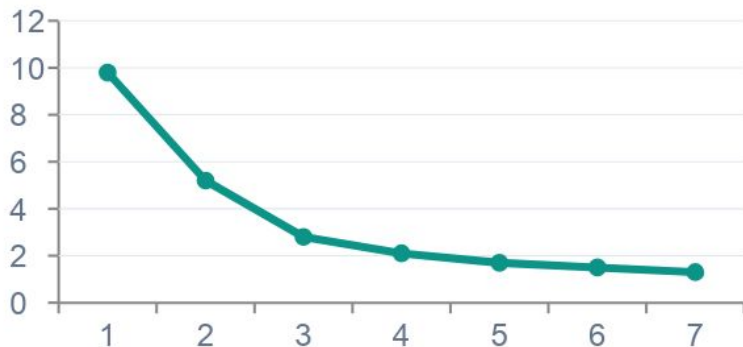
Distances inflate → clusters less meaningful.

Fix: ALWAYS run PCA before K-means on high-D data

How to Choose the Number of Clusters (k)?

Method 1: Elbow Plot

- Run K-means for $k = 1, 2, 3 \dots N$
- Plot inertia vs k
- Look for the 'elbow' — where gain flattens
 - Beware: not always obvious!



Method 2: Silhouette Score

- Measures how similar a point is to its own cluster vs others
- Score ranges: -1 (wrong cluster) to $+1$ (well separated)
- Choose k that maximises average silhouette
 - Works even when elbow is ambiguous

$$s = (b - a) / \max(a, b)$$

a = avg dist to same cluster
 b = avg dist to nearest other cluster



Hands-On: Real Brain Data

Real Example: Motor Task Brain States — ORIGAMI Lab



The Dataset

- fMRI recorded during a motor task from healthy subjects
- Each time point → one dynamic functional connectivity (dFC) feature vector
- dFC captures whole-brain region-to-region interaction patterns at each moment in time
- Each time point has a label: REST or TASK



The Question

Can K-means ($k=2$) on dFC time points recover REST vs TASK without ever seeing the labels?

Your notebook steps:

1

Load dFC features and labels (REST=0, TASK=1)

2

Visualise with PCA (2D) — do states already separate?

3

Run K-Means with $k=2$

4

Plot cluster assignments on the PCA space vs true labels

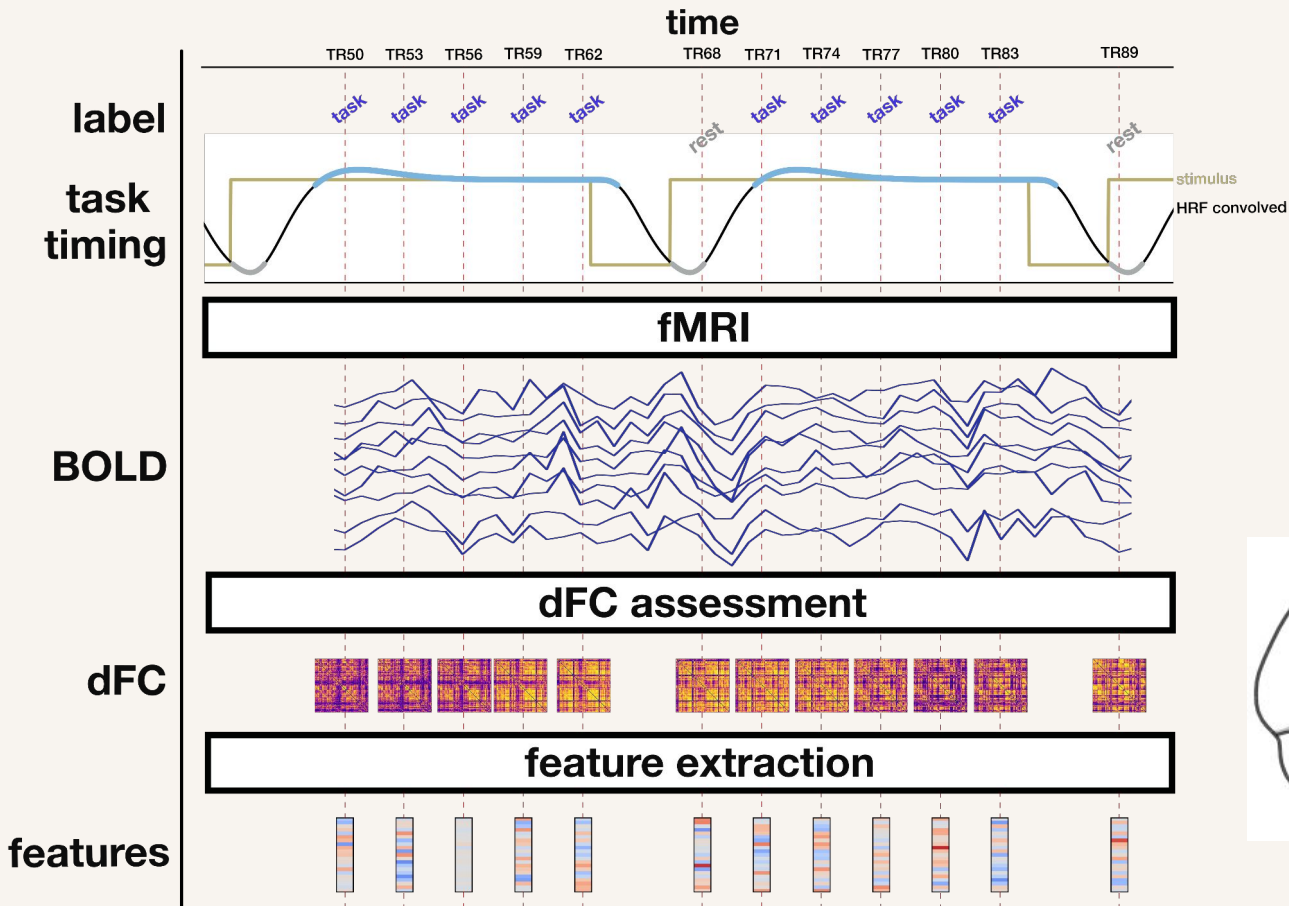
5

Compute Silhouette Score (internal) and ARI (external)

6

Interpret: what does a low/high ARI really mean here?

Real Example: Motor Task Brain States — ORIGAMI Lab



scikit-learn Cheatsheet for Today

1. Standardise features

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_train)
```

2. PCA → reduce dimensions

```
from sklearn.decomposition import PCA
pca = PCA(n_components=20)
X_pca = pca.fit_transform(X_scaled)
```

3. K-Means clustering (k=2: REST vs TASK)

```
from sklearn.cluster import KMeans
km = KMeans(n_clusters=2, n_init=10, random_state=42)
labels = km.fit_predict(X_pca)
```

4. Evaluation metrics

Coming soon ...



Evaluating Clusters

Two Types of Cluster Evaluation



Internal Validation

No true labels needed

- Silhouette Coefficient — how tight are clusters, how separated?
- Inertia / Within-cluster SS — lower = tighter (not comparable across k)
- Davies-Bouldin Index — ratio of within:between distances



External Validation

Requires known ground-truth labels

- Adjusted Rand Index (ARI) — compares label agreement; 0 = random, 1 = perfect
- Normalized Mutual Information (NMI) — information overlap between clusterings
- Adjusted Mutual Information — NMI corrected for chance

Internal Validation: Silhouette Score

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$$

$a(i)$ = mean distance to all points in same cluster | $b(i)$ = mean distance to nearest other cluster

Interpretation

$\approx +1 \rightarrow$ Well clustered

$\approx 0 \rightarrow$ On cluster boundary

$\approx -1 \rightarrow$ Wrong cluster

- Average silhouette over all points $\rightarrow$ single score for a given k
- Higher average silhouette = better clustering (use this to compare k values)
- Can be computed without any ground-truth labels
- **But: a high silhouette doesn't mean the clusters are biologically meaningful!**

External Validation: Adjusted Rand Index (ARI)

Intuition

- Consider all pairs of points
- Are they in the same cluster? Are they in the same true class?
- ARI measures how often clustering agrees with true labels — corrected for chance

Range & meaning

ARI = 1.0 → Perfect agreement

ARI = 0.0 → Random (no better than chance)

ARI = < 0 → Worse than random



Why external validation is critical in biology

- Internal metrics only tell you about the geometry of your clusters — not whether they are real
- In neuroscience: site effects, batch effects, age or sex confounds can create clusters that look great internally but mean nothing biologically
- External validation (e.g. ARI vs diagnosis, site, age group) links your clusters back to real-world meaning



Continue the exercise!

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3. K-Means clustering (k=2: REST vs TASK)

```
from sklearn.cluster import KMeans
km = KMeans(n_clusters=2, n_init=10, random_state=42)
labels = km.fit_predict(X_pca)
```

4. Evaluation metrics

```
from sklearn.metrics import silhouette_score,
adjusted_rand_score
si = silhouette_score(X_pca, labels)
ari = adjusted_rand_score(y_true, labels)
```

The ORIGAMI Result: What Do Our Numbers Tell Us?



K-Means (k=2) on dFC time points: Silhouette = 0.5 | ARI vs REST/TASK = 0.26

Silhouette = 0.5: not bad!

- The algorithm did find some structure in the dFC data
- Time points do separate into geometrically distinct groups
- But 2D PCA shows heavy overlap — the full picture is in higher dimensions

ARI = 0.26: that structure $\neq$ REST/TASK

- Clusters do not cleanly map onto the task design
- Subjects may not have been in the 'expected' brain state the whole time
- Motion artifacts, physiological noise, or mind-wandering could be driving the structure found

"Clustering always finds structure. *External validation tells us whether that structure is the one we care about.*"

If Not REST vs TASK — Then What?

Clustering found structure, but $ARI = 0.26$. So what did it find? Possibilities:



Motion states

Head movement during scanning creates artefactual connectivity patterns.
High-motion vs low-motion time points may cluster more strongly than REST vs TASK.



Attention & mind-wandering

Subjects may switch in and out of task engagement.
The 'TASK' label doesn't guarantee the subject was actually engaged.



Physiological fluctuations

Breathing and heart rate create global fMRI signal fluctuations that can dominate dFC patterns, regardless of task.



Intrinsic brain dynamics

The brain cycles through resting-state networks even during tasks.
These natural transitions may be stronger than the task effect.

→ *Each of these is a testable hypothesis: compute ARI against motion, ARI against attention ratings, etc.*

"If We Have Labels, Isn't Clustering Useless?"

✗ Common myth: "If we have REST/TASK labels, why bother clustering?"



Labels are often incomplete

REST/TASK is a coarse experimental label.
Clustering may reveal sub-states within each — e.g. early vs late task engagement, or different motor strategies.



Labels can be wrong or noisy

Was the subject really resting? Really engaged?
Clustering tests whether the data itself agrees with your labels — it is a sanity check on your design.



Clustering can extend labels

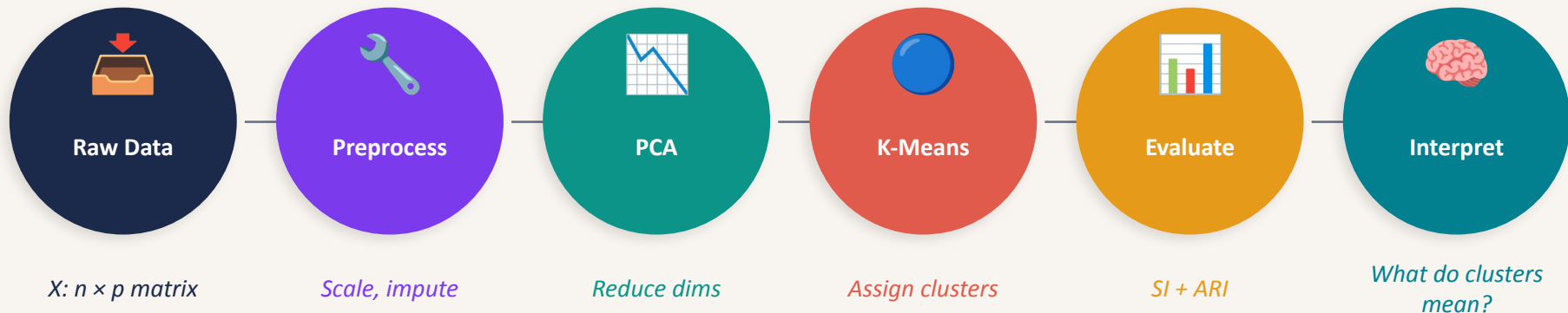
You use clustering to test, question, or enrich your labels — not replace them.
If clusters $\neq$ labels, that is informative, not a failure.

"Clustering answers: *What structure exists?*" External validation answers: *Do we care about that structure?*"



Putting It All Together

The Complete Clustering Pipeline



```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans

pipeline = make_pipeline(StandardScaler(), PCA(n_components=20), KMeans(n_clusters=2, n_init=10))
cluster_labels = pipeline.fit_predict(X_dfc) # X_dfc: time points x connectivity features
```

Key Take-Aways

- 1 Clustering = unsupervised grouping by similarity. No labels needed — but labels are still useful.
- 2 Always run PCA before K-means on high-dimensional data (dFC, genomics, imaging...).
- 3 Silhouette score tells you about cluster geometry — not whether the structure is biologically meaningful.
- 4 ARI (vs known labels) is the key external check: does the structure we found match what we care about?
- 5 Low ARI $\neq$ failure. It means: the data has structure, just not the structure you expected. Investigate why.
- 6 Clustering tests, questions, and extends labels — it doesn't replace them.